

Cloud gap filling for continuous normalized difference snow index snow cover using the dynamic seasonally recurrent snow depletion pattern over a mountainous watershed

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ABSTRACT

Cloud cover in remotely sensed snow cover is a persistent issue limiting accurate observation of snow cover. Cloud gap filling (CGF) seeks to infill cloud covered pixels, yet methods have historically focused on binary snow detection. Continuous normalized differenced snow index (NDSI) snow cover provides more information about snow cover than binary detection and is a valuable variable for modeling snowmelt runoff. Our goal was to use the dynamic seasonally recurrent snow depletion pattern to produce cloud free continuous NDSI snow cover for the period of 2000–2020 in the Upper Snake River Basin, Wyoming. We test two approaches to CGF for continuous NDSI snow cover, a one-to-one approach where one model is derived for each NDSI snow cover value, and a one-to-many approach where one model is derived to CGF all NDSI snow cover values. We then compared CGF results with the Moderate Resolution Imaging Spectroradiometer (MODIS) CGF product. We demonstrate the dynamic seasonally recurrent snow depletion pattern produces continuous daily NDSI snow cover with 96.23% accuracy over the NDSI snow cover range of 10–50 as calculated against independent cloud free (<10% cloud covered area) images from 2018 to 2020. Our model estimates of snow coverage were up to 10.8% lower than the MODIS CGF product. Accurate daily imagery of snow extent is essential for snowmelt modelers to forecast runoff and manage water resources because up to 1/6th of the world's population relies on water from snowmelt.

1. Introduction

Snow cover is an important component of the global energy budget (Moody et al., 2007), and remote sensing of snow is essential to monitor long term trends in mountainous regions (Bormann et al., 2018). In the Western United States, snowmelt serves as the primary source of freshwater with estimates as high as 50–80% (Stewart et al., 2004). Remotely sensed snow cover can supplement ground based data and can increase snowmelt runoff modeling skill (Bennett et al., 2019; Steele et al., 2017). Unfortunately, cloud cover, illumination, view angle, topography, and land cover all add uncertainty to remotely sensed snow cover (Dong, 2018; Riggs et al., 2017). Cloud cover is a persistent issue that can be addressed through cloud gap filling (CGF). CGF has thus far focused on binary mapping of snow or snow free, and few methods have been developed to produce continuous Normalized Difference Snow Index

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(NDSI) snow cover estimates. Despite this global importance, the high temporal variability of snowmelt requires daily observations which is currently hindered by persistent cloud cover.

Multiple methods of CGF exist (Li et al., 2019); however, nearly all methods produce maps of binary snow cover (Dong and Menzel, 2016; Gafurov et al., 2016; Hao et al., 2022; Li et al., 2017). Continuous NDSI estimates are essential for snowmelt runoff modeling because they can be directly converted to Fractional Snow Covered Area (fSCA), which is the percentage of snow cover within a pixel (Riggs et al., 2015). The continuous data captures the sub-pixel ablation and transitions that govern the timing and magnitude of melt that is oversimplified with binary snow maps. Consequently, snowmelt runoff modeling has improved accuracy and lower uncertainty when models use fSCA or even continuous NDSI snow cover instead of binary snow cover (Bennett et al., 2019; Di Marco et al., 2021; Siemens et al., 2021).

CGF can generally be broken into three categories: temporal, spatial, and multistep approaches. Temporal methods include a combination of the Moderate Resolution Imaging Spectroradiometer (MODIS) sensors on the Terra and Aqua satellites which have different pass over times (Parajka and Blöschl, 2008), temporal filtering by using a multi-day window to infill a cloudy pixel (Hori et al., 2017; Morriss et al., 2016; Parajka et al., 2012), and infilling with only the adjacent day observations (Dietz et al., 2013). Spatial methods like the “snowline method” achieve cloud removal by defining elevations of snow cover (Da Ronco and De Michele, 2014; Dietz et al., 2013; Parajka et al., 2010; Paudel and Andersen, 2011), or using adjacent cloud free pixels (Gafurov and Bárdossy, 2009; López-Burgos et al., 2013; Parajka and Blöschl, 2008; Paudel and Andersen, 2011). Multi-step processes leverage a combination of temporal and spatial steps to achieve up to 100% cloud cover removal on a daily time step (Da Ronco and De Michele, 2014; Gafurov et al., 2016; Gafurov and Bárdossy, 2009; Li et al., 2020). While these CGF approaches produce binary snow maps, the industry standard MOD10A1F uses a temporal approach to produce daily continuous NDSI estimates (Hall et al., 2019). However, MOD10A1F has a crucial limitation as it performs CGF by bringing the last observation forward. During periods of long cloud persistence, this reliance on temporal persistence may result in estimates that do not accurately represent the true snow state of the pixel (Gao et al., 2010; Hall et al., 2019). The temporal uncertainty inherent to the MOD10A1F CGF method creates a need for methods that do not rely on the persistence of previous observations.

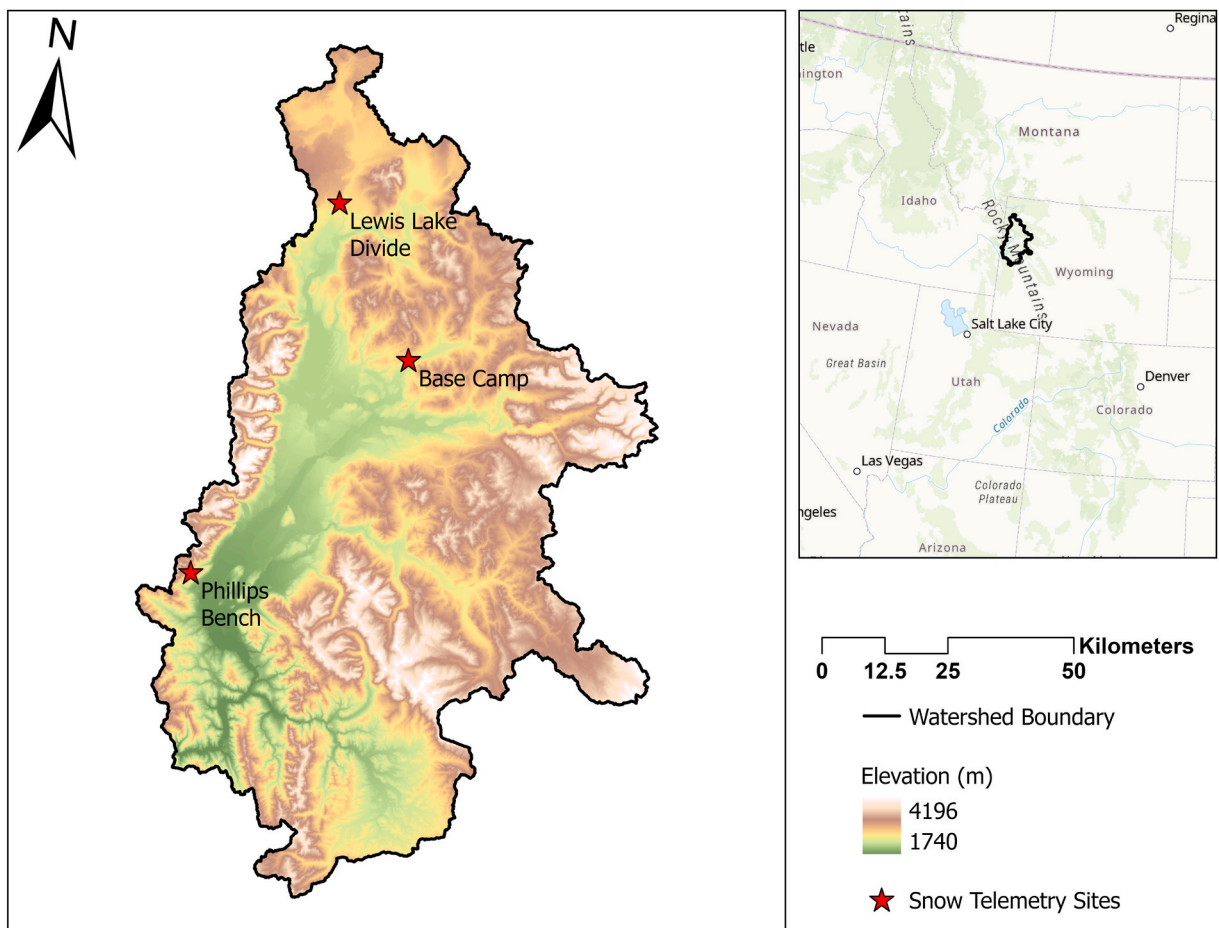


Fig. 1. The digital elevation model which ranged from 1740 to 4196 m for the Upper Snake River Basin is mapped along with the contextual study site location as an inset map.

Patterns of snow depletion exist seasonally, and are effective cloud removal methods that rely on spatio-temporal mathematical CGF, but it is unclear whether pattern based methods can produce continuous CGF NDSI snow cover. Binary snow cover mapping with CGF has been demonstrated to be accurate and highly similar even during severe drought (Woodruff et al., 2026; Woodruff and Qualls, 2019). Snow depletion patterns are applicable over a variety of landscapes making a pattern based approach enticing (Adams, 1976; König and Sturm, 1998; Luce and Tarboton, 2004; Parr et al., 2020; Pflug and Lundquist, 2020; Sturm and Wagner, 2010; Wang and Xie, 2009). CGF with a pattern-based approach relies on spatial relationships, thereby removing the cloud persistence uncertainties with temporal gap filling.

CGF methods have thus far focused on producing maps of binary snow cover rather than continuous snow cover estimates. Therefore, the goal of this study is to CGF and produce continuous NDSI snow cover with the dynamic seasonally recurrent snow depletion pattern. The objectives of this study are to a) develop candidate models of the dynamic seasonally recurrent snow depletion pattern to CGF; b) assess candidate model accuracy against cloud free daily imagery to determine the most accurate model; and c) compare the resulting CGF estimates with the freely available daily CGF snow cover product from the National Snow and Ice Data Center.

2. Study site and data source

2.1. Study site

The Upper Snake River Basin (Fig. 1) is a large watershed covering 8894 square kilometers. The Snake River is an important and highly regulated source of freshwater with multiple uses including agriculture, power generation, municipal water demands, and in-stream fish requirements (Qualls et al., 2013). Freshwater from the Upper Snake River Basin is dependent on winter snowpack. The downstream regulation of freshwater relies on accurate snowmelt runoff models. Improved snow cover datasets greatly aid modeling and regulation of snowmelt runoff making the Upper Snake River Basin an ideal study site to test new cloud gap filling approaches. A wide variety of land cover types are present in the watershed including sagebrush steppe (45.4% of the total area) and forested (47.9% of the total area) according to the 2015 National Land Cover Database (Homer et al., 2015). Elevations range from 1737 to 4194 m above mean sea level. The watershed receives complete snow coverage every year. During our study period, a range in melt out timing was observed. Melt out dates or the date snow water equivalent first reached a value of 0 cm at Base Camp, Lewis Lake Divide, and Phillips Bench ranged from as early as May first, May thirteenth, and May nineteenth, respectively to as late as June tenth, July third, and July sixth respectively (Table 1) (United States US Department of Agriculture, Natural Resource Conservation Service, n.d.). We also noted a large range in maximum snow water equivalent values during the study period with the greatest value of 144.5 cm recorded at Lewis Lake divide in 2011, and the minimum value of 22.1 cm at Base Camp in 2001.

2.2. Data source

We used the MODIS data collected from the Terra satellite for our study because the critical snow detection band six had a failure on the Aqua satellite (Shen et al., 2014). Even though a new algorithm was developed for the Aqua satellite, the MODIS sensor on board

Table 1
Melt out day and maximum snow water equivalent in centimeters by year for Base Camp, Lewis Lake Divide, and Phillips Bench snow telemetry sites.

Year	Base Camp Melt Out Date	Lewis Lake Divide Melt Out Date	Phillips Bench Melt Out Date	Base Camp Maximum Snow Water Equivalent (cm)	Lewis Lake Divide Maximum Snow Water Equivalent (cm)	Phillips Bench Maximum Snow Water Equivalent (cm)
2000	5/2	5/29	6/2	45.0	67.8	60.7
2001	5/1	5/18	5/26	22.1	48.5	47.8
2002	5/14	6/7	6/3	36.1	80.5	61.5
2003	5/18	6/5	6/2	53.1	87.6	64.5
2004	5/2	6/5	6/4	45.2	76.7	61.0
2005	5/9	5/27	6/16	32.5	62.5	58.7
2006	5/15	6/11	6/7	53.3	115.6	92.5
2007	5/1	5/19	5/19	35.3	61.5	47.8
2008	5/31	6/23	6/25	53.8	101.9	93.7
2009	5/19	6/12	6/14	41.1	95.0	79.8
2010	5/16	6/11	6/12	26.2	50.8	48.5
2011	6/10	7/3	7/6	66.8	144.5	111.8
2012	5/10	6/7	6/2	49.8	85.3	61.7
2013	5/13	5/28	6/3	39.4	79.8	64.8
2014	5/25	6/10	6/14	64.0	109.5	93.5
2015	5/4	5/13	6/2	38.1	56.6	55.4
2016	5/8	5/29	6/3	41.1	75.7	56.6
2017	5/30	6/18	6/19	68.8	123.7	106.7
2018	5/16	6/10	6/8	50.3	115.8	84.3
2019	5/21	6/15	6/17	47.8	101.6	81.3
2020	5/20	6/4	6/13	46.5	75.2	81.8

Aqua has more snow commission errors than the Terra MODIS sensor (Hall et al., 2019). We used daily data for the period of 2000–2020. The version six daily data contain NDSI snow cover (ranges from 0 to 100), cloud cover, ice, and quality assurance flags. We used the version 6 cloud containing dataset, which is maintained in the newer version 6.1 dataset, because we begin our modeling with a data preprocessing step that requires non cloud gap filled data. Our data preprocessing step did not include any cloud filtering or QA flag handling. We focused our study window on the melting period or Day of Year (DOY) 1–250, where DOY 1 is January 1 or maximum snow coverage at the watershed scale. We reserved 2000–2016 for training and 2017–2020 for testing. Our testing dataset included a range in melt out timing and snow water equivalent (Table 1). We downloaded cloud containing (MOD10A1) data for 2000–2020 and CGF data (MOD10A1F) for 2018–2020 for our study (Hall and Riggs, 2020). We compared our CGF results with the MOD10A1F CGF as a benchmark for CGF accuracy.

3. Methods

There are two logical approaches to remove cloud cover for continuous NDSI snow cover using the recurrent dynamic pattern of snow depletion: 1) develop a model for each individual value of NDSI snow cover and use it to remove cloud cover of the equivalent NDSI snow cover (a one-to-one approach), or 2) develop one model and remove cloud cover of all NDSI snow cover values (a one-to-many approach). We tested both approaches. We developed a model for each possible NDSI snow cover value from 10 to 90 in increments of 5 with the daily data from 2000 to 2016. The NDSI snow cover value is denoted by the subscript c throughout the remainder of the text. We selected intervals of 5 based on empirical observation of the range of NDSI snow cover values and their time series transition from 100 to 0. We noted that an interval of 5 captured the time series well while reducing computational costs for our analysis.

First, we developed the candidate models for each value of c . We then calculated accuracy for the one-to-one approach for cloud free images (<10% cloud cover 2000–2017). For the one-to-many assessment we selected the most accurate candidate model found in the one-to-one assessment. We then compared the one-to-many accuracy versus the direct one-to-one accuracies for cloud free imagery (<10% cloud cover) for 2018–2020. Our two-step assessment determines the best selection of c and the best approach: one-to-one or one-to-many. Candidate model derivation followed the same set of steps and only the value of c was altered. Once the most accurate approach was determined, we directly compared the output of our CGF process with the MOD10A1F CGF data product for 2018–2020.

3.1. Candidate model derivation

3.1.1. Data preparation

The dynamic seasonally recurrent pattern of snow depletion is contained in the time-series of MODIS daily data (Woodruff et al., 2026; Woodruff and Qualls, 2019). Seasonal composites are developed that record the Last Day Snow (LDS) was recorded for a pixel and the First Day Land (FDL) was recorded for each pixel. Beginning on DOY 1 we iterate through every available MODIS image for a year and record the day of LDS and FDL and save these dates to rasters. The FDL and LDS yearly maps indicate the first day where a pixel is observed to change from a snow to snow-free state. The FDL and LDS composites record the first instance of snow disappearance, ignoring secondary snowfall. The recurring pattern of melt is observed during the first state change and subsequent post-melt snowfall represents stochastic processes that differ from year to year. We calculated FDL and LDS for each year 2000–2016. The DOY of the LDS and FDL is dependent on the NDSI snow cover threshold c . For the lowest threshold, $c = 10$, we calculated and LDS and FDL where snow covered pixels were defined as those with c -values greater 10 and snow free pixels had c less than 10. We repeated this procedure and developed FDL and LDS images for each year (2000–2016) for each value of c (10 through 90 at intervals of 5). We use a subscript c to denote each FDL and LDS image based on the threshold used to develop it, e.g., FDL_c . A total of 289 FDLs were calculated or 17 years multiplied by 17 values of c . Woodruff and Qualls (2019) demonstrated either the FDL or the LDS could be used to derive a model of the dynamic seasonally recurrent snow depletion pattern with essentially the same pattern. We present maps of select FDL_c composites and qualitatively investigated the time series of daily MODIS imagery.

3.1.2. Candidate model derivation

The dynamic seasonally recurrent snow depletion pattern can be derived by combining multiple years of FDL_c composites with the same value of c with a Principal Component Analysis (PCA) (Woodruff et al., 2026; Woodruff and Qualls, 2019). PCA accomplished two important steps: the multiple seasons are combined into a single principal component (the recurrent snow depletion pattern was contained in the first principal component), and noise-like cloud cover that may influence the FDL_c composites was discarded in subsequent principal components. Just as we calculated a FDL_c for each value of c , we also calculated a candidate model for each value of c , referred to as PCA_c . For each year and a constant value of c , we transferred the FDL_c rasters into vector with length N . Once all FDL_c rasters were converted to vectors, they were combined into a M by N matrix where M is equal to the number of years (2000–2016) and N is equal to the number of MODIS pixels contained in the FDL_c Equation (1). We maintained the centroid x and y coordinates of each pixel in separate columns of the M by N matrix. PCA was applied to the N by M matrix. We then apply matrix multiplication of the first principal component (eigen vector) to create the PCA_c values (t), Equation (1). The resulting PCA_c values were then reattached to the x and y coordinates and rasterized to a spatial raster equal in spatial resolution and geolocation to the daily MODIS imagery. Equation (1) was applied for each value of c resulting in 17 PCA_c models. The process follows a t -mode PCA that is commonly used in atmospheric models and the critical transformation of a t -mode PCA is applied to produce a matrix of date and order. The t -mode PCA is used for identifying consistent spatial patterns over time – clustering times with an output of spatial pattern – which is preferred over an s -mode PCA which identifies regions with similar temporal behavior – clustering locations with an output of spatial importance. In

the case of the dynamic seasonally recurrent snow depletion pattern each pixel's melt order is produced. The result of the t-mode PCA for the raster model is an ordered map of each pixel's melt. Melt order indicates the temporal progression of melt across space (raster pixels) of seasonal snow depletion. We report variance described by the first principal components (PCA_c) and the factor loadings or correlation of each FDL_c to the resulting PCA_c (Jensen, 2016).

$$\begin{bmatrix} 1, 1 & \dots & 1, M \\ \vdots & \ddots & \vdots \\ N, 1 & \dots & N, M \end{bmatrix} * \begin{bmatrix} 1 \\ \vdots \\ M \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ N \end{bmatrix} \rightarrow \begin{bmatrix} t_1 & x_1 & y_1 \\ \vdots & \vdots & \vdots \\ t_N & x_N & y_N \end{bmatrix} \rightarrow \text{PCA}_c \text{ (Spatial Raster)} \quad (1)$$

3.2. Candidate model accuracy assessment

Candidate model accuracy assessment was conducted for the one-to-one and one-to-many modeling approaches. Each daily MODIS image (Img) was converted to a binary snow and snow free image by selecting the value of c , which differentiated snow from snow free. Similar to the FDL_c and PCA_c we denote c selection for the daily imagery as Img_c. For the one-to-one accuracy assessment, the PCA_c was directly compared to Img_c, where c was the same for both (e.g., PCA₁₀ compared to Img₁₀, and PCA₁₅ compared to Img₁₅). For the one-to-many accuracy assessment a PCA_c was compared to the range of Img_c values (e.g., PCA₁₀ compared with Img₁₀, Img₁₅, Img₂₀, etc.). Accuracy was assessed by the same metric for all of the candidate models and both approaches, one-to-one and one-to-many. We evaluated candidate model accuracy by first fitting the candidate model to Img_c and we calculated accuracy as the percentage of correctly classified pixels.

3.2.1. Candidate model image fitting

We fit a PCA_c candidate model to an Img_c by gradually altering a threshold (τ). The unique pixel values in a given PCA_c represent the order of pixel melt, from first pixel to melt to last. We iteratively increase τ for converting the unique PCA_c pixel values to snow covered (pixel values greater than τ) and snow free (pixel values less than τ). This creates a binary snow/no-snow model PCA_c(τ). As τ was increased the number of pixels considered snow free increased. We select the best fit τ , comparing the converted PCA_c(τ) to Img_c by calculating the visible pixel error: VPE (Equation (2)). The comparison of Img_c to PCA_c(τ) for each value of τ yielded three values: the number of incorrectly modeled snow free pixels (I_B), incorrectly modeled snow pixels (I_A), and the total number of pixels visible (T_P). The best fit selection of τ produced the minimum VPE. For each Img_c, τ was fit with only the information contained in the comparison of PCA_c to Img_c.

$$\text{VPE} = \sqrt{\frac{I_B^2 + I_A^2}{T_P^2}} \quad (2)$$

3.2.2. Candidate model versus cloud free image accuracy

Accuracy was calculated as the number of correctly modeled pixels, for each comparison of PCA_c versus a given Img_c. We calculated accuracy by equation (3) with the previously defined I_A , I_B , and T_P . Accuracy was assessed for each candidate model, and both approaches with all the cloud free images (<10% cloud cover) for the period of 2000-2017. We included 2017 in the candidate model assessment as an independent year to ensure unbiased selection of the best candidate model and approach for cloud gap filling.

$$\text{ACC} = \left(1 - \frac{I_B + I_A}{T_P}\right) \times 100\% \quad (3)$$

3.3. PCA_c cloud removal comparison with the CGF product

We compared CGF results for continuous NDSI snow cover from our best candidate model and approach, derived following the methods above, and the MOD10A1F CGF product for the period of 2018-2020. The MOD10A1F CGF product brings the last observed value for a pixel forward for cloud covered pixels. This approach achieves 100% cloud removal but inherently may overestimate snow cover during the melting period especially where cloud cover was persistent. Persistent cloud coverage adds uncertainty in a pixels NDSI snow cover value especially during snowmelt. With this CGF approach, cloud persistence for 7 days while a pixel is actively melting may greatly overestimate the true value of NDSI snow cover poorly representing the time series of a declining NDSI snow cover value. We selected the most accurate candidate model and approach (one-to-one or one-to-many) identified by following section 3.2. The PCA_c method is spatially based, and cloud removal requires some visible portion of the watershed to CGF, therefore we filtered out daily MODIS images with greater than 75% cloud covered area. We did not implement any bias corrections for the MOD10A1F dataset prior to comparison as we used the comparison to benchmark our CGF approach.

3.4. Production of continuous NDSI snow cover

We produced semi-continuous NDSI snow cover maps by compositing binary snow cover images for a range of c values into a single image representing an incrementally continuous NDSI CGF image, based on the increments of c used. For example, to produce a continuous NDSI snow cover image for value of c ranging from 10 to 90 at increments of 5, we generate binary snow cover images for each value of c . The composite image would then be generated from the 17 binary images, one for each value of c . The continuous NDSI

snow cover resolution could be increased by the user by selecting and processing finer increments of c to composite in daily semi-continuous NDSI snow cover images. We compared the maps of our continuous NDSI snow cover with the MOD10A1F images.

4. Results

4.1. PCA_c dynamic seasonal pattern candidate model derivation

We derived seasonal composites (FDL_c) for the range of c values 10–90 in increments of five. Fig. 2 shows the 2006 seasonal composite FDL_{10} , FDL_{40} and FDL_{90} in panels a, b, and c, respectively. The corresponding PCA_{10} , PCA_{40} , and PCA_{90} are given in Fig. 2 panels d, e, and f, respectively. In Fig. 2a, FDL_{10} depicts a smooth gradient over the watershed for the first to melt pixels to the last to melt pixels (dark to light). The corresponding PCA_{10} (Fig. 2d) also has a smooth gradient. As c increased to 40 (Fig. 2b–e), we observe a few dark pixels surrounded completely by light-colored pixels. These islands of dark pixels completely surrounded by light pixels can be interpreted as meaning the dark pixel melted far earlier than the neighboring lightly colored pixels. For FDL_{90} and PCA_{90} (Fig. 2c–f) the pixels are more similar to distinct categories rather than the continuous gradient we would expect of snow depletion. The behavior of higher c values could be the result lower NDSI snow cover sensitivity in forested environments where snow sloughs off tree branches leading to a lower NDSI snow cover value which is further complicated by viewing angle and topography.

We investigated the source of the differing FDL_c composites and PCA_c models in Fig. 2 by plotting the 2018 time series of MODIS images. Each daily image was converted to values of Img_{10} , Img_{25} , and Img_{40} (Fig. 3a) and Img_{10} , Img_{50} , Img_{70} and Img_{90} (Fig. 3b). Img_{10} , Img_{25} , and Img_{40} all exhibit a smooth transition from 100% to 0% snow covered area (SCA) between day of year 100 and 200 (Fig. 3a). In contrast, Img_{90} values are all near to 0% SCA (Fig. 3b, blue crosses). The time series suggests the temporal progression of snow depletion at larger values of c , e.g., 90, do not follow the same progression as lower values of c . From Figs. 2 and 3 for higher values of c , we might infer that the dynamic seasonally recurrent snow depletion pattern may not be as easily derived due to a less smooth temporal progression compared to lower values of c .

4.2. Candidate PCA_c performance

The candidate model factor loadings and variance for the direct comparisons, e.g., PCA_{10} versus FDL_{10} for all seasonal composites for 2000–2016 is presented in Table 2. As the value of c increases (Table 2, column 1) the variance generally decreases from 0.936 to 0.406. Minimum, average, and maximum factor loadings were also highest at the smallest value of c , PCA_{10} . The decline in variance

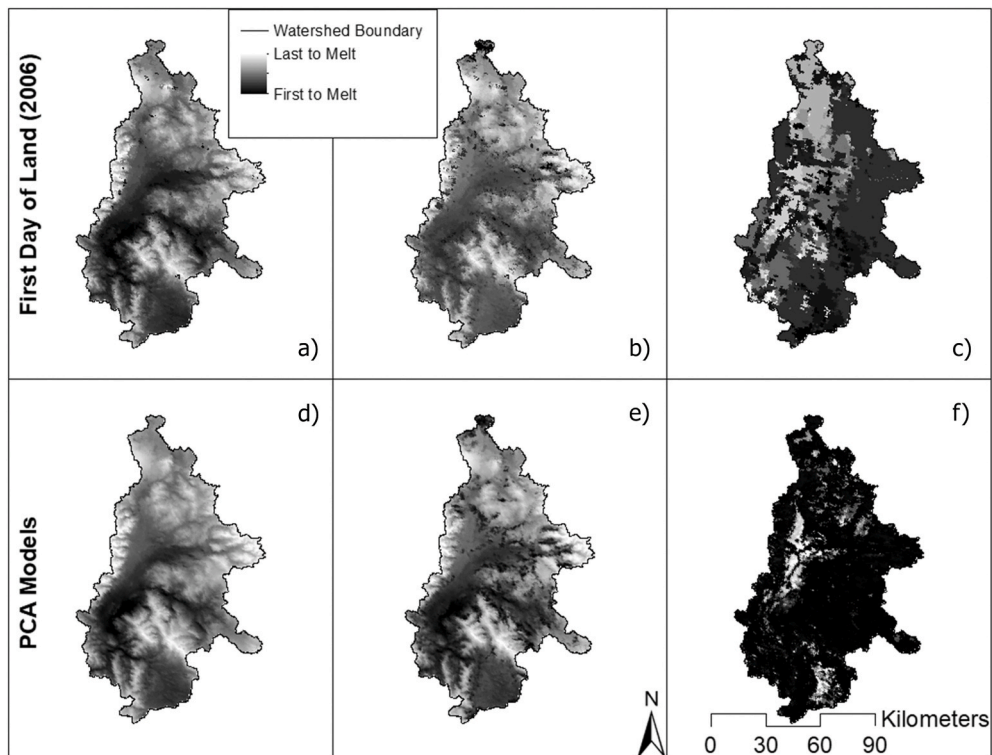


Fig. 2. FDL seasonal composites for 2006 are mapped with varied values of c in panels a, b, and c for FDL_{10} , FDL_{40} , and FDL_{90} , respectively. PCA models developed with FDL_c seasonal composites from 2000 to 2016 are mapped in panels d, e, and f for PCA_{10} , PCA_{40} , and PCA_{90} , respectively. In all six panels the melt order is shown in a gradient from dark to light or earliest to melt to last to melt.

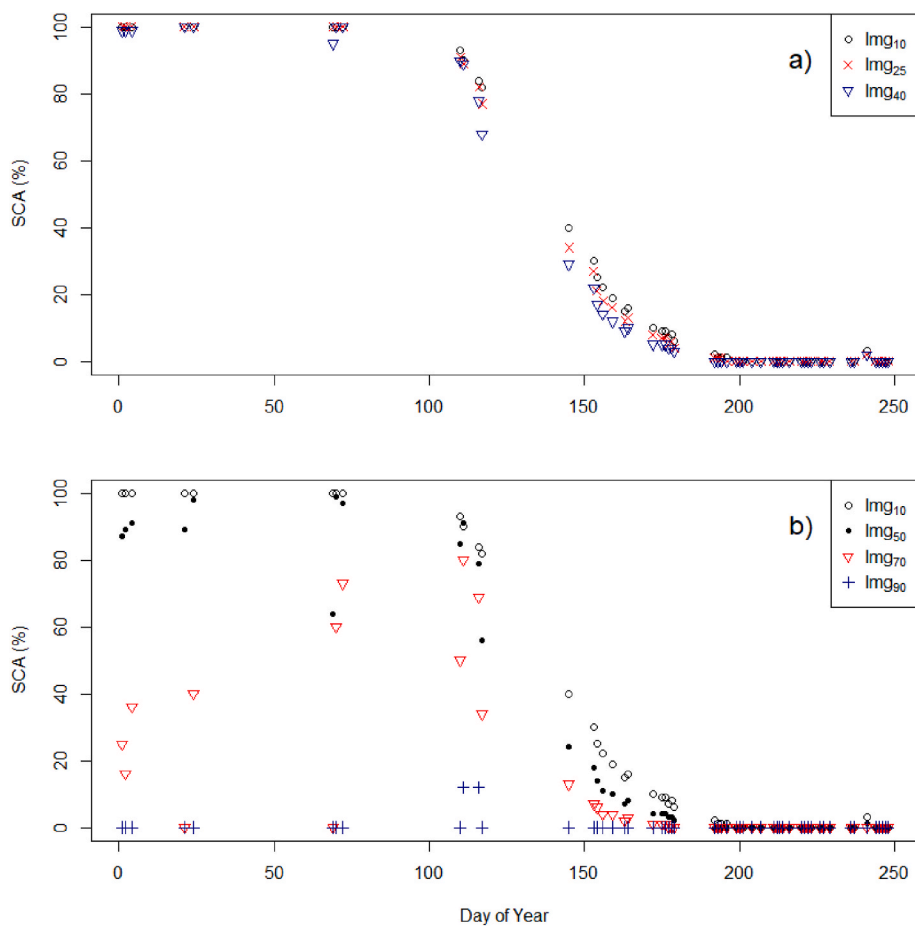


Fig. 3. Time series of daily SCA from MODIS imagery for 2018. In panel a, SCA values for Img_{10} , Img_{25} , and Img_{40} are given by the black hollow circles, red X's, and blue triangles, respectively. In panel b, SCA values for Img_{10} , Img_{50} , Img_{70} and Img_{90} are given by the black hollow circles, black solid circles, red triangles, and blue crosses, respectively.

Table 2

Descriptive statistics including variance and factor loading for each candidate PCA_c model are given. The total dataset variance described by each PCA_c model is given as well as the minimum, average, and maximum factor loadings for the comparison of each PCA_c versus the corresponding set of FDL_c seasonal composites (2000–2016).

PCA_c	First Principal Component Variance	Factor Loading Minimum	Factor Loading Average	Factor Loading Maximum
PCA_{10}	0.936	0.949	0.968	0.979
PCA_{15}	0.929	0.945	0.964	0.977
PCA_{20}	0.919	0.935	0.959	0.974
PCA_{25}	0.905	0.921	0.952	0.967
PCA_{30}	0.889	0.908	0.943	0.958
PCA_{35}	0.871	0.892	0.933	0.951
PCA_{40}	0.848	0.877	0.920	0.940
PCA_{45}	0.820	0.866	0.905	0.929
PCA_{50}	0.784	0.840	0.884	0.913
PCA_{55}	0.725	0.787	0.851	0.894
PCA_{60}	0.645	0.710	0.797	0.870
PCA_{65}	0.606	0.640	0.763	0.853
PCA_{70}	0.643	0.555	0.786	0.862
PCA_{75}	0.670	0.505	0.797	0.891
PCA_{80}	0.620	0.480	0.750	0.860
PCA_{85}	0.491	0.253	0.596	0.832
PCA_{90}	0.406	0.011	0.386	0.859

and factor loadings also aligns with the observed differences in the FDL_c seasonal composites (Fig. 2) and poor temporal signal at larger values of c (Fig. 3b).

4.3. One-to-one candidate model accuracy assessment

Boxplots of PCA_c candidate model accuracy for all cloud free (<10% cloud cover) imagery 2000–2017 are given in Fig. 4. The most accurate models have the shortest colored box and the most variable candidate models have the tallest colored boxes. PCA₁₀ through PCA₃₅ have similar box heights (Fig. 4). The box height begins to grow at PCA₄₀ (Fig. 4). Outliers (hollow circles) exist for all of the candidate models. Interestingly, at values of c greater than 60 the median accuracy (center line within each box) is greater than 99%. Overinterpretation of the high accuracies at larger values of c should be tempered due to concerns of inconsistent NDSI snow cover values with varied zenith angles and topography. Additionally, interpretations should be cautiously made because high median accuracy coincides with the categorical snow depletion of PCA₉₀, Fig. 2f. As depicted in Fig. 3b large values of c produce daily estimates of snow cover of 0%. High accuracy in Fig. 4, for large values of c are likely images that are considered completely snow free throughout the entire snow depletion season, and accuracy may be inflated. The lower limit of the box for PCA₁₀ through PCA₅₅ is fully above 90%, Fig. 4. This range of c values produces accurate results and has a smooth gradient describing pixel melt order (Fig. 2). PCA₁₀ is the most accurate (average accuracy 98.69%), and PCA₅₅ is comparable (average accuracy of 95.23%). For the one-to-one approach, candidate models PCA₁₀ through PCA₅₅ are reasonably comparable; however, the one-to-one approach has an obvious setback: multiple models are required to produce continuous cloud free NDSI snow cover: one for each value of c .

4.4. One-to-many candidate model accuracy assessment

PCA₁₀ is the most accurate one-to-one candidate model. We tested PCA₁₀ as the candidate model for the one-to-many approach for cloud free imagery between 2018 and 2020. Average accuracy of the one-to-one approach for a given value of c and the average accuracy improvement by year is shown in Table 3. We restricted the analysis to images between 100% and 1% SCA. The largest increase in average accuracy was recorded in 2019 at 2.42% for PCA₁₀ fit to Img₄₀ compared to the one-to-one PCA₄₀ fit to Img₄₀ at 2.42%. Accuracy improvements decline after a c of 40 and become negative at a c of 60. For the range of c between 10 and 50 average accuracies for the PCA₁₀ one-to-many approach were between 97.39% and 92.94% for the period of 2018–2020 with an overall accuracy of 96.23%. Importantly, significant gains in computational efficiency are realized with the one-to-many approach. FDL generation is the greatest computational requirement, and the one-to-many approach requires an FDL for only one value of c where the one-to-one approach requires an FDL_c for each PCA_c. For the values of c we chose, the one-to-many approach reduces computational requirements by a factor of 17.

4.5. Cloud removal performance compared between CGF products

The comparison of CGF results for the PCA₁₀ model (one-to-many approach) versus the MOD10A1F CGF product for images in 2018–2020 with less than 75% cloud covered area are plotted for four selections of c , Fig. 5. Each point in Fig. 5 represents an image where the PCA₁₀ SCA is plotted along the x axis and MOD10A1F CGF value is plotted along the y axis. By plotting SCA versus SCA, we can show all the CGF results for all 3 years on a single plot. Fig. 5a–d presents the SCA (%) where snow cover corresponds to values of c

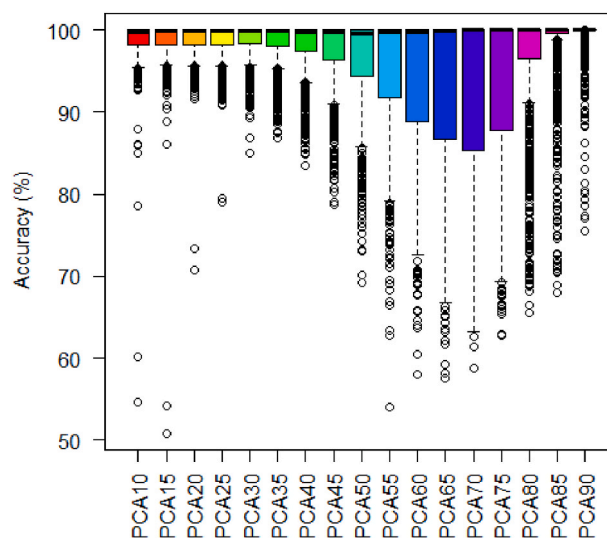


Fig. 4. Spatial accuracies (%) of candidate models versus cloud free images (2000–2017) are plotted as boxplots. Quantiles are shown by the box height, and the median is given as the black bar within the box. Outliers are plotted as hollow circles.

Table 3

Accuracies for images with less than 10% cloud cover and SCA between 100% and 1% are given for the one-to-one approach as well (columns 2-4) and accuracy improvements realized with the one-to-many approach (columns 5-7). Average accuracy is given as a percentage.

NDSI Snow Cover (c)	One-to-One Accuracy 2018	One-to-One Accuracy 2019	One-to-One Accuracy 2020	One-to-Many Accuracy Improvement 2018	One-to-Many Accuracy Improvement 2019	One-to-Many Accuracy Improvement 2020
10	97.07	97.33	95.25	0	0	0
15	96.91	97.24	95.90	0.10	0.19	0.12
20	96.77	97.08	97.50	0.28	0.42	0.12
25	96.56	96.58	97.19	0.48	0.74	0.18
30	96.27	96.01	96.78	0.67	1.22	0.27
35	95.81	95.09	96.28	1.04	1.91	0.43
40	95.22	93.71	94.53	1.42	2.42	1.16
45	93.60	93.43	93.58	1.46	1.70	1.29
50	91.66	92.20	91.32	0.68	1.34	1.62
55	89.49	89.13	88.14	0.36	0.75	1.32
60	88.29	85.20	84.17	-0.83	-4.36	-3.13
65	86.20	82.99	81.74	-2.27	-9.35	-7.89
70	83.78	81.02	80.75	-2.28	-11.64	-8.55
75	78.95	80.77	79.15	5.53	-11.26	-9.08
80	76.32	81.28	80.54	10.91	-13.66	-4.03
85	82.61	88.59	85.29	5.43	-3.15	0.92
90	83.36	92.32	92.21	4.95	-8.94	-0.58

of 10, 25, 40, and 55, respectively. The MOD10A1F CGF produces larger estimates of snow cover (points falling above the one-to-one line) for c values of 10, 25, and 40 (Fig. 5abc). The MOD10A1F CGF overestimation of snow cover is most notable between 20 and 70% SCA (x-axis, Fig. 5abc). The maximum estimation difference for a c of 10, Fig. 5a, was 10.8%. Similarly, at a c of 55 the MOD10A1F produced larger SCA estimates for x-axis SCA between 0% and about 50% (Fig. 5d). Conversely, for SCA above 50% the PCA₁₀ estimates began to exceed the MOD10A1F estimates, becoming more exaggerated for x axis SCA values above 80%.

We mapped continuous NDSI snow cover derived with the PCA₁₀ one-to-many approach for DOY 117 and 145 in 2018 which had

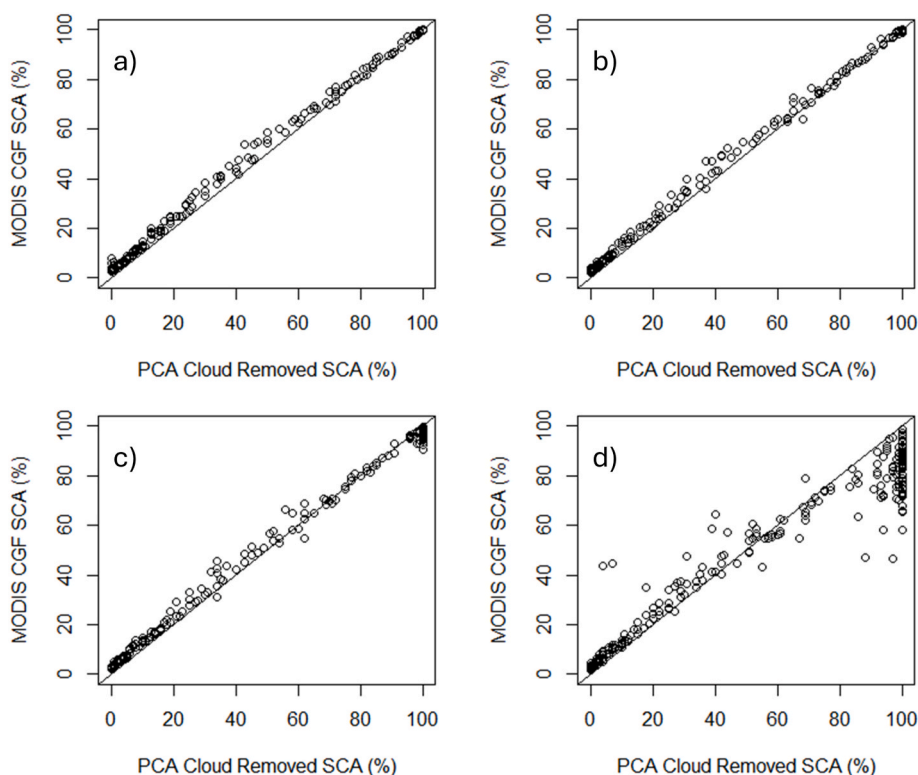


Fig. 5. CGF PCA₁₀ data is compared directly with the MOD10A1F CGF product. Daily MODIS cloud removed SCA values for days with <75% cloud cover for 2018-2020 are plotted. Four comparisons are shown: SCA where snow cover is determined as c > 10 (a), c > 25 (b), c > 40 (c), and c > 55 (d). Each point represents a single MODIS image where the x axis is the PCA10 estimate and the y-axis is the MOD10A1F estimate for the image. A one-to-one line is shown in each plot to indicate differences between CGF estimates.

82% and 40% SCA, respectively. The result is compared with the MOD10A1F CGF product in Fig. 6. NDSI snow cover is given as a gradient of blues, where dark blue is low NDSI snow cover and light blue is high, and snow free is brown. Based on the previous findings we set a threshold of 50, and all NDSI snow cover values greater than 50 were mapped as the same color. For DOY 117, the PCA₁₀ model produces a smoother and larger spatially distributed gradient of NDSI snow cover values (Fig. 6a) than the MOD10A1F product (Fig. 6c). Later in the melting season PCA₁₀ model still generated a smoother gradient (Fig. 6b) than the MOD10A1F product (Fig. 6d). NDSI snow covers tend to have a more spatially erratic distribution with the MOD10A1F product.

5. Discussion

Estimating daily snow cover requires rapid and accurate methods because seasonal snow is dynamic, covers large portions of the earth surface (13%) (Johnston et al., 2024), is essential for modeling snow melt runoff (Rango and Martinec, 1979), and significantly impacts global albedo (Hall, 2004). With the vulnerability of seasonal snow cover to a changing climate (Barnett et al., 2005), accurate daily estimation is important for long term monitoring and climatological studies (Vavrus, 2007). The dynamic seasonally recurrent snow depletion pattern is an efficient and accurate CGF method that relies only on cloud free pixels in a given image to perform CGF. The robust nature and recurrence of a snow depletion pattern is largely due to the major drivers of snow depletion that are unchanging year after year like elevation, slope, aspect, (Dingman, 2008; Hock, 2003). Patterns are similar during drought (Woodruff et al., 2026), and as demonstrated here can produce estimates of continuous NDSI snow cover with a purely spatial model.

Accurate CGF with continuous NDSI snow cover values have significant implications for snowline mapping. We demonstrated snow cover mapping average accuracy of 98.69% for binary snow detection of NDSI snow cover >10 which is similar to other studies (Gafurov et al., 2016; Li et al., 2017); however, unlike other studies we produced continuous NDSI snow cover for the range of 10-55 in increments of 5 with average accuracies equal to or greater than 95.23%. We found that MOD10A1F produced larger CGF SCA values than the PCA method for all NDSI snow cover values from c of 10 to 50 owing to the differences between their respective SCA CGF methods. Because MOD10A1F performs CGF by bringing the last observation forward, it likely tends to overestimate snow extent on a daily basis because snow melt is generally moving from snow covered to snow free. This is problematic because errors near the snowline, which may be overestimated by MOD10A1F, negatively impact the accuracy of physical melt processes. Comparing CGF by MOD10A1F with the PCA method, MOD10A1F generally produces larger NDSI snow cover values at a given location. Larger NDSI snow cover value approximately corresponds to pushing the snowline elevation lower in the watershed, highlighting the importance of accurate daily estimates.

Physical snowmelt modeling requires accurate snowline representation. For example, latent heat exchanges near the snowline can represent 33% of the total snowmelt energy (Harder et al., 2017). As snow melts it may become patchy, and patchy snow has a longer snowline and larger melt contributions from sensible and latent heat advection (Harder et al., 2017; Mott et al., 2011; Schlögl et al., 2018). Patchy snow would be indicated by a lower NDSI snow cover value. Another explanation for lower or inconsistent NDSI snow cover values may be the result of angular effects for different viewing angles (Wang et al., 2022). The dynamic seasonally recurrent snow depletion pattern CGF is completed based on visible pixels across a large area, some of which do not experience the angular effects, potentially mitigating any CGF inconsistencies that would be present with MOD10A1F or other temporal CGF approaches.

6. Conclusions

In this study we produced continuous NDSI CGF with a truly spatially method that doesn't require the use of retrospective cloud free observations like MOD10A1F. We demonstrated the one-to-many approach with the PCA₁₀ model could achieve accurate continuous NDSI snow cover CGF for a range of NDSI snow cover. Average accuracy was 96.23% for the NDSI snow covers between 10 and 50 when compared to cloud free (<10% cloud cover) daily images (2018-2020). When our CGF was compared to the temporal gap filling MOD10A1F product, we found the daily snow extent was higher for the MOD10A1F product by 10.8%. Future research should investigate the accuracy of the dynamic seasonally recurrent pattern of snow depletion for CGF over a range of climactic regions and assess the resulting CGF product for snowmelt runoff modeling. We demonstrate for the first time that spatial patterns in seasonal snow depletion are an observable watershed-scale characteristic that are recurrent and accurate over a wide range of NDSI snow cover values. The daily CGF methodology presented is applicable to gauged and ungauged basins and provides decision makers with the accurate, timely, and cloud free data necessary to manage water resources from snowmelt runoff in the Western United States and similar systems worldwide.

CRediT authorship contribution statement

C.D. Woodruff: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **R.J. Qualls:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **K.S. Humes:** Writing – review & editing, Visualization, Investigation, Funding acquisition.

Ethical Statement

Hereby, I Craig Woodruff consciously assure that for the manuscript *Cloud Gap Filling for Continuous Normalized Difference Snow Index Snow Cover using the Dynamic Seasonally Recurrent Snow Depletion Pattern over a Mountainous Watershed* the following is fulfilled:

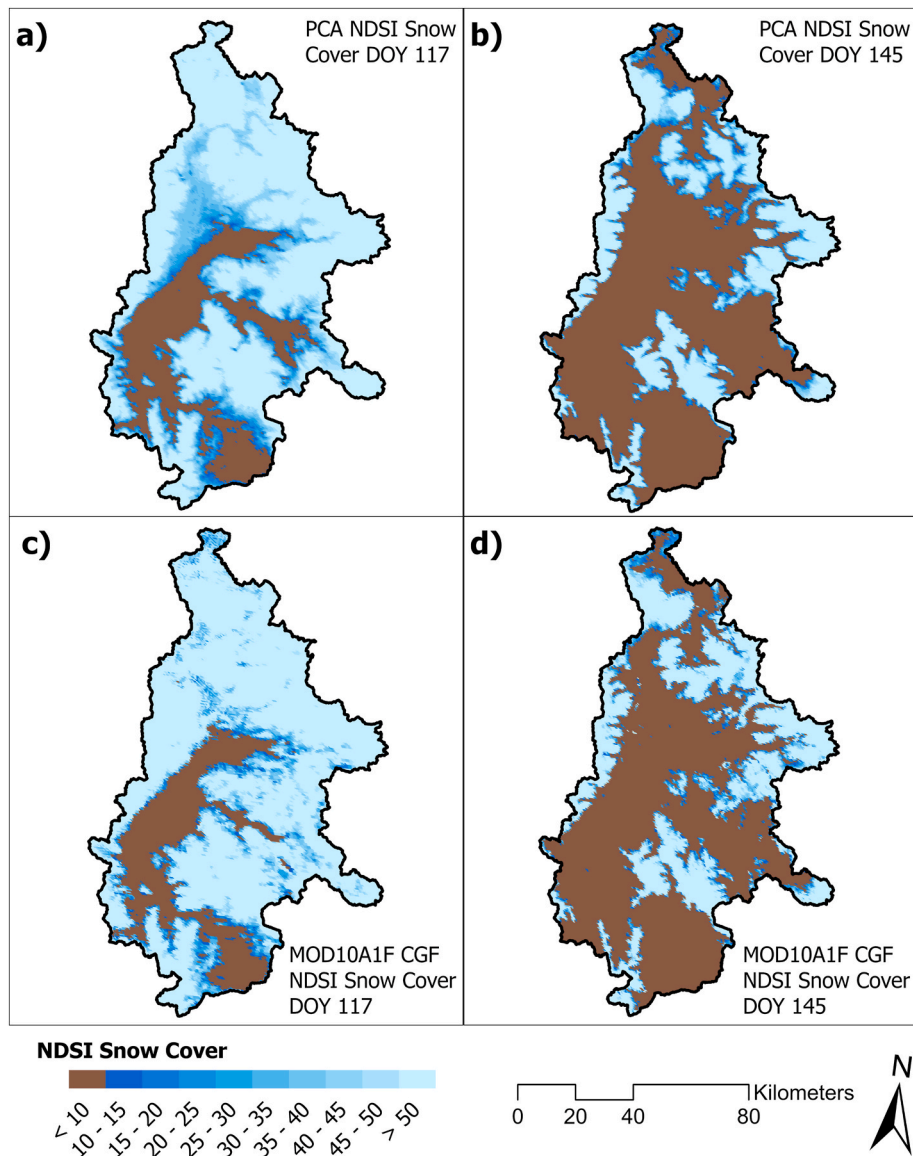


Fig. 6. Comparison of continuous NDSI snow cover maps generated with the PCA₁₀ method for DOY 117 (a) and DOY 145 (b) in 2018 and the MOD10A1F CGF NDSI snow cover for DOY 117 (c) and DOY 145 (d). The gradient of blue from dark to light indicate increasing NDSI snow cover and brown indicates snow free.

- 1) This material is the authors' own original work, which has not been previously published elsewhere.
- 2) The paper is not currently being considered for publication elsewhere.
- 3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
- 4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
- 5) The results are appropriately placed in the context of prior and existing research.
- 6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.
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The violation of the Ethical Statement rules may result in severe consequences.

I agree with the above statements and declare that this submission follows the policies of *Remote Sensing Applications: Society and Environment* as outlined in the Guide for Authors and in the Ethical Statement.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Glossary

CGF	Cloud Gap Filling
NDSI	Normalized Differenced Snow Index
MODIS	Moderate Resolution Imaging Spectroradiometer
fSCA	fractional Snow-Covered Area
NSIDC	National Snow and Ice Data Center
LDS	Last Day of Snow
FDL:	First Day of Land
PCA	Principal Component Analysis
Img	a MODIS image clipped to the watershed boundary
VPE	Visible Pixel Error
SCA	Snow Covered Area
VIIRS	Visible Infrared Imaging Radiometer Suite
DOY	Day of Year

Data availability

Research Link Provided.

[MOD10A1 and MOD10A1F \(Original data\)](#) (National Snow and Ice Data Center)

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